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01

what you can't say

january 2004
paulgraham.com/say.html

what can't we say? one way to find these ideas is simply to look at things people do say, and get in trouble for.[2]

of course, we're not just looking for things we can't say. we're looking for things we can't say that are true, or at least have enough chance of being true that the question should remain open. but many of the things people get in trouble for saying probably do make it over this second, lower threshold. no one gets in trouble for saying that 2 + 2 is 5, or that people in pittsburgh are ten feet tall. such obviously false statements might be treated as jokes, or at worst as evidence of insanity, but they are not likely to make anyone mad. the statements that make people mad are the ones they worry might be believed. i suspect the statements that make people maddest are those they worry might be true.

if galileo had said that people in padua were ten feet tall, he would have been regarded as a harmless eccentric. saying the earth orbited the sun was another matter. the church knew this would set people thinking.

certainly, as we look back on the past, this rule of thumb works well. a lot of the statements people got in trouble for seem harmless now. so it's likely that visitors from the future would agree with at least some of the statements that get people in trouble today. do we have no galileos? not likely.

to find them, keep track of opinions that get people in trouble, and start asking, could this be true? ok, it may be heretical (or whatever modern equivalent), but might it also be true?

heresy

this won't get us all the answers, though. what if no one happens to have gotten in trouble for a particular idea yet? what if some idea would be so radioactively controversial that no one would dare express it in public? how can we find these too?

another approach is to follow that word, heresy. in every period of history, there seem to have been labels that got applied to statements to shoot them down before anyone had a chance to ask if they were true or not. "blasphemy", "sacrilege", and "heresy" were such labels for a good part of western history, as in more recent times "indecent", "improper", and "unamerican" have been. by now these labels have lost their sting. they always do. by now they're mostly used ironically. but in their time, they had real force.

the word "defeatist", for example, has no particular political connotations now. but in germany in 1917 it was a weapon, used by ludendorff in a purge of those who favored a negotiated peace. at the start of world war ii it was used extensively by churchill and his supporters to silence their opponents. in 1940, any argument against churchill's aggressive policy was "defeatist". was it right or wrong? ideally, no one got far enough to ask that.

we have such labels today, of course, quite a lot of them, from the all-purpose "inappropriate" to the dreaded "divisive." in any period, it should be easy to figure out what such labels are, simply by looking at what people call ideas they disagree with besides untrue. when a politician says his opponent is mistaken, that's a straightforward criticism, but when he attacks a statement as "divisive" or "racially insensitive" instead of arguing that it's false, we should start paying attention.

so another way to figure out which of our taboos future generations will laugh at is to start with the labels. take a label — "sexist", for example — and try to think of some ideas that would be called that. then for each ask, might this be true?

just start listing ideas at random? yes, because they won't really be random. the ideas that come to mind first will be the most plausible ones. they'll be things you've already noticed but didn't let yourself

you can see this on a small scale in the matter of dirty words. a lot of my friends are starting to have children now, and they're all trying not to use words like "fuck" and "shit" within baby's hearing, lest baby start using these words too. but these words are part of the language, and adults use them all the time. so parents are giving their kids an inaccurate idea of the language by not using them. why do they do this? because they don't think it's fitting that kids should use the whole language. we like children to seem innocent.[7]

most adults, likewise, deliberately give kids a misleading view of the world. one of the most obvious examples is santa claus. we think it's cute for little kids to believe in santa claus. i myself think it's cute for little kids to believe in santa claus. but one wonders, do we tell them this stuff for their sake, or for ours?

i'm not arguing for or against this idea here. it is probably inevitable that parents should want to dress up their kids' minds in cute little baby outfits. i'll probably do it myself. the important thing for our purposes is that, as a result, a well brought-up teenage kid's brain is a more or less complete collection of all our taboos — and in mint condition, because they're untainted by experience. whatever we think that will later turn out to be ridiculous, it's almost certainly inside that head.

how do we get at these ideas? by the following thought experiment. imagine a kind of latter-day conrad character who has worked for a time as a mercenary in africa, for a time as a doctor in nepal, for a time as the manager of a nightclub in miami. the specifics don't matter — just someone who has seen a lot. now imagine comparing what's inside this guy's head with what's inside the head of a well-behaved sixteen year old girl from the suburbs. what does he think that would shock her? he knows the world; she knows, or at least embodies, present taboos. subtract one from the other, and the result is what we can't say.

mechanism

i can think of one more way to figure out what we can't say: to look at how taboos are created. how do moral fashions arise, and why are they adopted? if we can understand this mechanism, we may be able to see it at work in our own time.

moral fashions don't seem to be created the way ordinary fashions are. ordinary fashions seem to arise by accident when everyone imitates the whim of some influential person. the fashion for broad-toed shoes in late fifteenth century europe began because charles viii of france had six toes on one foot. the fashion for the name gary began when the actor frank cooper adopted the name of a tough mill town in indiana. moral fashions more often seem to be created deliberately. when there's something we can't say, it's often because some group doesn't want us to.

the prohibition will be strongest when the group is nervous. the irony of galileo's situation was that he got in trouble for repeating copernicus's ideas. copernicus himself didn't. in fact, copernicus was a canon of a cathedral, and dedicated his book to the pope. but by galileo's time the church was in the throes of the counter-reformation and was much more worried about unorthodox ideas.

to launch a taboo, a group has to be poised halfway between weakness and power. a confident group doesn't need taboos to protect it. it's not considered improper to make disparaging remarks about americans, or the english. and yet a group has to be powerful enough to enforce a taboo. coprophiles, as of this writing, don't seem to be numerous or energetic enough to have had their interests promoted to a lifestyle.

i suspect the biggest source of moral taboos will turn out to be power struggles in which one side only barely has the upper hand. that's where you'll find a group powerful enough to enforce taboos, but

in the sciences, especially, it's a great advantage to be able to question assumptions. the m.o. of scientists, or at least of the good ones, is precisely that: look for places where conventional wisdom is broken, and then try to pry apart the cracks and see what's underneath. that's where new theories come from.

a good scientist, in other words, does not merely ignore conventional wisdom, but makes a special effort to break it. scientists go looking for trouble. this should be the m.o. of any scholar, but scientists seem much more willing to look under rocks.[10]

why? it could be that the scientists are simply smarter; most physicists could, if necessary, make it through a phd program in french literature, but few professors of french literature could make it through a phd program in physics. or it could be because it's clearer in the sciences whether theories are true or false, and this makes scientists bolder. (or it could be that, because it's clearer in the sciences whether theories are true or false, you have to be smart to get jobs as a scientist, rather than just a good politician.)

whatever the reason, there seems a clear correlation between intelligence and willingness to consider shocking ideas. this isn't just because smart people actively work to find holes in conventional thinking. i think conventions also have less hold over them to start with. you can see that in the way they dress.

it's not only in the sciences that heresy pays off. in any competitive field, you can win big by seeing things that others daren't. and in every field there are probably heresies few dare utter. within the us car industry there is a lot of hand-wringing now about declining market share. yet the cause is so obvious that any observant outsider could explain it in a second: they make bad cars. and they have for so long that by now the us car brands are antibrands — something you'd buy a car despite, not because of. cadillac stopped being the cadillac of cars in about 1970. and yet i suspect no one dares say this.[11] otherwise these companies would have tried to fix the problem.

training yourself to think unthinkable thoughts has advantages beyond the thoughts themselves. it's like stretching. when you stretch before running, you put your body into positions much more extreme than any it will assume during the run. if you can think things so outside the box that they'd make people's hair stand on end, you'll have no trouble with the small trips outside the box that people call innovative.

pensieri stretti

when you find something you can't say, what do you do with it? my advice is, don't say it. or at least, pick your battles.

suppose in the future there is a movement to ban the color yellow. proposals to paint anything yellow are denounced as "yellowist", as is anyone suspected of liking the color. people who like orange are tolerated but viewed with suspicion. suppose you realize there is nothing wrong with yellow. if you go around saying this, you'll be denounced as a yellowist too, and you'll find yourself having a lot of arguments with anti-yellowists. if your aim in life is to rehabilitate the color yellow, that may be what you want. but if you're mostly interested in other questions, being labelled as a yellowist will just be a distraction. argue with idiots, and you become an idiot.

the most important thing is to be able to think what you want, not to say what you want. and if you feel you have to say everything you think, it may inhibit you from thinking improper thoughts. i think it's better to follow the opposite policy. draw a sharp line between your thoughts and your speech. inside your head, anything is allowed. within my head i make a point of encouraging the most outrag-

best of all, probably, is humor. zealots, whatever their cause, invariably lack a sense of humor. they can't reply in kind to jokes. they're as unhappy on the territory of humor as a mounted knight on a skating rink. victorian prudishness, for example, seems to have been defeated mainly by treating it as a joke. likewise its reincarnation as political correctness. "i am glad that i managed to write 'the crucible,'" arthur miller wrote, "but looking back i have often wished i'd had the temperament to do an absurd comedy, which is what the situation deserved."[17]

abq

a dutch friend says i should use holland as an example of a tolerant society. it's true they have a long tradition of comparative open-mindedness. for centuries the low countries were the place to go to say things you couldn't say anywhere else, and this helped to make the region a center of scholarship and industry (which have been closely tied for longer than most people realize). descartes, though claimed by the french, did much of his thinking in holland.

and yet, i wonder. the dutch seem to live their lives up to their necks in rules and regulations. there's so much you can't do there; is there really nothing you can't say?

certainly the fact that they value open-mindedness is no guarantee. who thinks they're not open-minded? our hypothetical prim miss from the suburbs thinks she's open-minded. hasn't she been taught to be? ask anyone, and they'll say the same thing: they're pretty open-minded, though they draw the line at things that are really wrong. (some tribes may avoid "wrong" as judgemental, and may instead use a more neutral sounding euphemism like "negative" or "destructive".)

when people are bad at math, they know it, because they get the wrong answers on tests. but when people are bad at open-mindedness they don't know it. in fact they tend to think the opposite. remember, it's the nature of fashion to be invisible. it wouldn't work otherwise. fashion doesn't seem like fashion to someone in the grip of it. it just seems like the right thing to do. it's only by looking from a distance that we see oscillations in people's idea of the right thing to do, and can identify them as fashions.

time gives us such distance for free. indeed, the arrival of new fashions makes old fashions easy to see, because they seem so ridiculous by contrast. from one end of a pendulum's swing, the other end seems especially far away.

to see fashion in your own time, though, requires a conscious effort. without time to give you distance, you have to create distance yourself. instead of being part of the mob, stand as far away from it as you can and watch what it's doing. and pay especially close attention whenever an idea is being suppressed. web filters for children and employees often ban sites containing pornography, violence, and hate speech. what counts as pornography and violence? and what, exactly, is "hate speech?" this sounds like a phrase out of *1984*.

labels like that are probably the biggest external clue. if a statement is false, that's the worst thing you can say about it. you don't need to say that it's heretical. and if it isn't false, it shouldn't be suppressed. so when you see statements being attacked as x-ist or y-ic (substitute your current values of x and y), whether in 1630 or 2030, that's a sure sign that something is wrong. when you hear such labels being used, ask why.

especially if you hear yourself using them. it's not just the mob you need to learn to watch from a distance. you need to be able to watch your own thoughts from a distance. that's not a radical idea, by the way; it's the main difference between children and adults. when a child gets angry because he's tired, he doesn't know what's happening. an adult can distance himself enough from the situation to say

02

inequality and risk

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paulgraham.com/inequality.html

our startup paid its first round of outside investors 36x. which meant, with current us tax rates, that it made sense to invest in us if we had better than a 1 in 24 chance of succeeding. that sounds about right. that's probably roughly how we looked when we were a couple of nerds with no business experience operating out of an apartment.

if that kind of risk doesn't pay, venture investing, as we know it, doesn't happen.

that might be ok if there were other sources of capital for new companies. why not just have the government, or some large almost-government organization like fannie mae, do the venture investing instead of private funds?

i'll tell you why that wouldn't work. because then you're asking government or almost-government employees to do the one thing they are least able to do: take risks.

as anyone who has worked for the government knows, the important thing is not to make the right choices, but to make choices that can be justified later if they fail. if there is a safe option, that's the one a bureaucrat will choose. but that is exactly the wrong way to do venture investing. the nature of the business means that you want to make terribly risky choices, if the upside looks good enough.

vcs are currently paid in a way that makes them focus on the upside: they get a percentage of the fund's gains. and that helps overcome their understandable fear of investing in a company run by nerds who look like (and perhaps are) college students.

if vcs weren't allowed to get rich, they'd behave like bureaucrats. without hope of gain, they'd have only fear of loss. and so they'd make the wrong choices. they'd turn down the nerds in favor of the smooth-talking mba in a suit, because that investment would be easier to justify later if it failed.

founders

but even if you could somehow redesign venture funding to work without allowing vcs to become rich, there's another kind of investor you simply cannot replace: the startups' founders and early employees.

what they invest is their time and ideas. but these are equivalent to money; the proof is that investors are willing (if forced) to treat them as interchangeable, granting the same status to "sweat equity" and the equity they've purchased with cash.

the fact that you're investing time doesn't change the relationship between risk and reward. if you're going to invest your time in something with a small chance of succeeding, you'll only do it if there is a proportionately large payoff.[2] if large payoffs aren't allowed, you may as well play it safe.

like many startup founders, i did it to get rich. but not because i wanted to buy expensive things. what i wanted was security. i wanted to make enough money that i didn't have to worry about money. if i'd been forbidden to make enough from a startup to do this, i would have sought security by some other means: for example, by going to work for a big, stable organization from which it would be hard to get fired. instead of busting my ass in a startup, i would have tried to get a nice, low-stress job at a big research lab, or tenure at a university.

that's what everyone does in societies where risk isn't rewarded. if you can't ensure your own security, the next best thing is to make a nest for yourself in some large organization where your status depends mostly on seniority.[3]

even if we could somehow replace investors, i don't see how we could replace founders. investors mainly contribute money, which in principle is the same no matter what the source. but the founders contribute ideas. you can't replace those.

discover has already been invented elsewhere. and the only thing you can offer in return is raw materials and cheap labor. once you sink that low, other countries can do whatever they like with you: install puppet governments, siphon off your best workers, use your women as prostitutes, dump their toxic waste on your territory — all the things we do to poor countries now. the only defense is to isolate yourself, as communist countries did in the twentieth century. but the problem then is, you have to become a police state to enforce it.

wealth and power

i realize startups are not the main target of those who want to eliminate economic inequality. what they really dislike is the sort of wealth that becomes self-perpetuating through an alliance with power. for example, construction firms that fund politicians' campaigns in return for government contracts, or rich parents who get their children into good colleges by sending them to expensive schools designed for that purpose. but if you try to attack this type of wealth through *economic* policy, it's hard to hit without destroying startups as collateral damage.

the problem here is not wealth, but corruption. so why not go after corruption?

we don't need to prevent people from being rich if we can prevent wealth from translating into power. and there has been progress on that front. before he died of drink in 1925, commodore vanderbilt's wastrel grandson reggie ran down pedestrians on five separate occasions, killing two of them. by 1969, when ted kennedy drove off the bridge at chappaquiddick, the limit seemed to be down to one. today it may well be zero. but what's changed is not variation in wealth. what's changed is the ability to translate wealth into power.

how do you break the connection between wealth and power? demand transparency. watch closely how power is exercised, and demand an account of how decisions are made. why aren't all police interrogations videotaped? why did 36% of princeton's class of 2007 come from prep schools, when only 1.7% of american kids attend them? why did the us really invade iraq? why don't government officials disclose more about their finances, and why only during their term of office?

a friend of mine who knows a lot about computer security says the single most important step is to log everything. back when he was a kid trying to break into computers, what worried him most was the idea of leaving a trail. he was more inconvenienced by the need to avoid that than by any obstacle deliberately put in his path.

like all illicit connections, the connection between wealth and power flourishes in secret. expose all transactions, and you will greatly reduce it. log everything. that's a strategy that already seems to be working, and it doesn't have the side effect of making your whole country poor.

i don't think many people realize there is a connection between economic inequality and risk. i didn't fully grasp it till recently. i'd known for years of course that if one didn't score in a startup, the other alternative was to get a cozy, tenured research job. but i didn't understand the equation governing my behavior. likewise, it's obvious empirically that a country that doesn't let people get rich is headed for disaster, whether it's diocletian's rome or harold wilson's britain. but i did not till recently understand the role risk played.

if you try to attack wealth, you end up nailing risk as well, and with it growth. if we want a fairer world, i think we're better off attacking one step downstream, where wealth turns into power.

03

economic inequality

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paulgraham.com/ineq.html

even people sophisticated enough to know about the pie fallacy are led toward it by the custom of describing economic inequality as a ratio of one quantile's income or wealth to another's. it's so easy to slip from talking about income shifting from one quantile to another, as a figure of speech, into believing that is literally what's happening.

except in the degenerate case, economic inequality can't be described by a ratio or even a curve. in the general case it consists of multiple ways people become poor, and multiple ways people become rich. which means to understand economic inequality in a country, you have to go find individual people who are poor or rich and figure out why. [2]

if you want to understand *change* in economic inequality, you should ask what those people would have done when it was different. this is one way i know the rich aren't all getting richer simply from some new system for transferring wealth to them from everyone else. when you use the would-have method with startup founders, you find what most would have done back in 1960, when economic inequality was lower, was to join big companies or become professors. before mark zuckerberg started facebook, his default expectation was that he'd end up working at microsoft. the reason he and most other startup founders are richer than they would have been in the mid 20th century is not because of some right turn the country took during the reagan administration, but because progress in technology has made it much easier to start a new company that grows fast.

traditional economists seem strangely averse to studying individual humans. it seems to be a rule with them that everything has to start with statistics. so they give you very precise numbers about variation in wealth and income, then follow it with the most naive speculation about the underlying causes.

but while there are a lot of people who get rich through rent-seeking of various forms, and a lot who get rich by playing zero-sum games, there are also a significant number who get rich by creating wealth. and creating wealth, as a source of economic inequality, is different from taking it — not just morally, but also practically, in the sense that it is harder to eradicate. one reason is that variation in productivity is accelerating. the rate at which individuals can create wealth depends on the technology available to them, and that grows exponentially. the other reason creating wealth is such a tenacious source of inequality is that it can expand to accommodate a lot of people.

—

i'm all for shutting down the crooked ways to get rich. but that won't eliminate great variations in wealth, because as long as you leave open the option of getting rich by creating wealth, people who want to get rich will do that instead.

most people who get rich tend to be fairly driven. whatever their other flaws, laziness is usually not one of them. suppose new policies make it hard to make a fortune in finance. does it seem plausible that the people who currently go into finance to make their fortunes will continue to do so, but be content to work for ordinary salaries? the reason they go into finance is not because they love finance but because they want to get rich. if the only way left to get rich is to start startups, they'll start startups. they'll do well at it too, because determination is the main factor in the success of a startup. [3] and while it would probably be a good thing for the world if people who wanted to get rich switched from playing zero-sum games to creating wealth, that would not only not eliminate great variations in wealth, but might even exacerbate them. in a zero-sum game there is at least a limit to the upside. plus a lot of the new startups would create new technology that further accelerated variation in productivity.

notice how novel it feels to think about that. the public conversation so far has been exclusively about the need to decrease economic inequality. we've barely given a thought to how to live with it.

i'm hopeful we'll be able to. brandeis was a product of the gilded age, and things have changed since then. it's harder to hide wrongdoing now. and to get rich now you don't have to buy politicians the way railroad or oil magnates did. [6] the great concentrations of wealth i see around me in silicon valley don't seem to be destroying democracy.

there are lots of things wrong with the us that have economic inequality as a symptom. we should fix those things. in the process we may decrease economic inequality. but we can't start from the symptom and hope to fix the underlying causes. [7]

the most obvious is poverty. i'm sure most of those who want to decrease economic inequality want to do it mainly to help the poor, not to hurt the rich. [8] indeed, a good number are merely being sloppy by speaking of decreasing economic inequality when what they mean is decreasing poverty. but this is a situation where it would be good to be precise about what we want. poverty and economic inequality are not identical. when the city is turning off your water because you can't pay the bill, it doesn't make any difference what larry page's net worth is compared to yours. he might only be a few times richer than you, and it would still be just as much of a problem that your water was getting turned off.

closely related to poverty is lack of social mobility. i've seen this myself: you don't have to grow up rich or even upper middle class to get rich as a startup founder, but few successful founders grew up desperately poor. but again, the problem here is not simply economic inequality. there is an enormous difference in wealth between the household larry page grew up in and that of a successful startup founder, but that didn't prevent him from joining their ranks. it's not economic inequality per se that's blocking social mobility, but some specific combination of things that go wrong when kids grow up sufficiently poor.

one of the most important principles in silicon valley is that "you make what you measure." it means that if you pick some number to focus on, it will tend to improve, but that you have to choose the right number, because only the one you choose will improve; another that seems conceptually adjacent might not. for example, if you're a university president and you decide to focus on graduation rates, then you'll improve graduation rates. but only graduation rates, not how much students learn. students could learn less, if to improve graduation rates you made classes easier.

economic inequality is sufficiently far from identical with the various problems that have it as a symptom that we'll probably only hit whichever of the two we aim at. if we aim at economic inequality, we won't fix these problems. so i say let's aim at the problems.

for example, let's attack poverty, and if necessary damage wealth in the process. that's much more likely to work than attacking wealth in the hope that you will thereby fix poverty. [9] and if there are people getting rich by tricking consumers or lobbying the government for anti-competitive regulations or tax loopholes, then let's stop them. not because it's causing economic inequality, but because it's stealing. [10]

if all you have is statistics, it seems like that's what you need to fix. but behind a broad statistical measure like economic inequality there are some things that are good and some that are bad, some that are historical trends with immense momentum and others that are random accidents. if we want to fix the world behind the statistics, we have to understand it, and focus our efforts where they'll do the most good.

between government receipts as a percentage of gdp, which have remained more or less constant during the entire period from the end of world war ii to the present, and tax rates, which have varied dramatically.

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note: this is a new version from which i removed a pair of metaphors that made a lot of people mad, essentially by macroexpanding them. if anyone wants to see the old version, i put it here.

related:

04

modeling a wealth tax

august 2020
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may 2026
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free" in scare quotes when talking about returns, the kind of risks you're talking about now are the almost apocalyptic kind that would make tax rates a moot point.

2 the same conversion rate applies to capital gains. the source of the multiple is whether the money is taxed every year or just once. indeed it's the same math you'd use to calculate the value of any income-generating asset.

3 you can deduct some state tax from your federal income taxes, but there's a cap on how much you can deduct, which means in the marginal case we simply add the two rates.

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06

the refragmentation

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paulgraham.com/re.html

professional managers. economies of scale ruled the day. it seemed to people at the time that this was the final state of things. john d. rockefeller said in 1880

> the day of combination is here to stay. individualism has gone, never to return.

he turned out to be mistaken, but he seemed right for the next hundred years.

the consolidation that began in the late 19th century continued for most of the 20th. by the end of world war ii, as michael lind writes, "the major sectors of the economy were either organized as government-backed cartels or dominated by a few oligopolistic corporations."

for consumers this new world meant the same choices everywhere, but only a few of them. when i grew up there were only 2 or 3 of most things, and since they were all aiming at the middle of the market there wasn't much to differentiate them.

one of the most important instances of this phenomenon was in tv. here there were 3 choices: nbc, cbs, and abc. plus public tv for eggheads and communists. the programs that the 3 networks offered were indistinguishable. in fact, here there was a triple pressure toward the center. if one show did try something daring, local affiliates in conservative markets would make them stop. plus since tvs were expensive, whole families watched the same shows together, so they had to be suitable for everyone.

and not only did everyone get the same thing, they got it at the same time. it's difficult to imagine now, but every night tens of millions of families would sit down together in front of their tv set watching the same show, at the same time, as their next door neighbors. what happens now with the super bowl used to happen every night. we were literally in sync. [6]

in a way mid-century tv culture was good. the view it gave of the world was like you'd find in a children's book, and it probably had something of the effect that (parents hope) children's books have in making people behave better. but, like children's books, tv was also misleading. dangerously misleading, for adults. in his autobiography, robert macneil talks of seeing gruesome images that had just come in from vietnam and thinking, we can't show these to families while they're having dinner.

i know how pervasive the common culture was, because i tried to opt out of it, and it was practically impossible to find alternatives. when i was 13 i realized, more from internal evidence than any outside source, that the ideas we were being fed on tv were crap, and i stopped watching it. [7] but it wasn't just tv. it seemed like everything around me was crap. the politicians all saying the same things, the consumer brands making almost identical products with different labels stuck on to indicate how prestigious they were meant to be, the balloon-frame houses with fake "colonial" skins, the cars with several feet of gratuitous metal on each end that started to fall apart after a couple years, the "red delicious" apples that were red but only nominally apples. and in retrospect, it *was* crap. [8]

but when i went looking for alternatives to fill this void, i found practically nothing. there was no internet then. the only place to look was in the chain bookstore in our local shopping mall. [9] there i found a copy of *the atlantic*. i wish i could say it became a gateway into a wider world, but in fact i found it boring and incomprehensible. like a kid tasting whisky for the first time and pretending to like it, i preserved that magazine as carefully as if it had been a book. i'm sure i still have it somewhere. but though it was evidence that there was, somewhere, a world that wasn't red delicious, i didn't find it till college.

it wasn't just as consumers that the big companies made us similar. they did as employers too. within companies there were powerful forces pushing people toward a single model of how to look and act.

starting one's own company seemed too ambitious, but because it didn't seem ambitious enough. even as late as the 1970s, when i grew up, the ambitious plan was to get lots of education at prestigious institutions, and then join some other prestigious institution and work one's way up the hierarchy. your prestige was the prestige of the institution you belonged to. people did start their own businesses of course, but educated people rarely did, because in those days there was practically zero concept of starting what we now call a startup: a business that starts small and grows big. that was much harder to do in the mid 20th century. starting one's own business meant starting a business that would start small and stay small. which in those days of big companies often meant scurrying around trying to avoid being trampled by elephants. it was more prestigious to be one of the executive class riding the elephant.

by the 1970s, no one stopped to wonder where the big prestigious companies had come from in the first place. it seemed like they'd always been there, like the chemical elements. and indeed, there was a double wall between ambitious kids in the 20th century and the origins of the big companies. many of the big companies were roll-ups that didn't have clear founders. and when they did, the founders didn't seem like us. nearly all of them had been uneducated, in the sense of not having been to college. they were what shakespeare called rude mechanicals. college trained one to be a member of the professional classes. its graduates didn't expect to do the sort of grubby menial work that andrew carnegie or henry ford started out doing. [15]

and in the 20th century there were more and more college graduates. they increased from about 2% of the population in 1900 to about 25% in 2000. in the middle of the century our two big forces intersect, in the form of the gi bill, which sent 2.2 million world war ii veterans to college. few thought of it in these terms, but the result of making college the canonical path for the ambitious was a world in which it was socially acceptable to work for henry ford, but not to be henry ford. [16]

i remember this world well. i came of age just as it was starting to break up. in my childhood it was still dominant. not quite so dominant as it had been. we could see from old tv shows and yearbooks and the way adults acted that people in the 1950s and 60s had been even more conformist than us. the mid-century model was already starting to get old. but that was not how we saw it at the time. we would at most have said that one could be a bit more daring in 1975 than 1965. and indeed, things hadn't changed much yet.

but change was coming soon. and when the duplo economy started to disintegrate, it disintegrated in several different ways at once. vertically integrated companies literally dis-integrated because it was more efficient to. incumbents faced new competitors as (a) markets went global and (b) technical innovation started to trump economies of scale, turning size from an asset into a liability. smaller companies were increasingly able to survive as formerly narrow channels to consumers broadened. markets themselves started to change faster, as whole new categories of products appeared. and last but not least, the federal government, which had previously smiled upon j. p. morgan's world as the natural state of things, began to realize it wasn't the last word after all.

what j. p. morgan was to the horizontal axis, henry ford was to the vertical. he wanted to do everything himself. the giant plant he built at river rouge between 1917 and 1928 literally took in iron ore at one end and sent cars out the other. 100,000 people worked there. at the time it seemed the future. but that is not how car companies operate today. now much of the design and manufacturing happens in a long supply chain, whose products the car companies ultimately assemble and sell. the reason car companies operate this way is that it works better. each company in the supply chain focuses on what they know best. and they each have to do it well or they can be swapped out for another supplier.

version 1 of the national economy consisted of a few big blocks whose relationships were negotiated in back rooms by a handful of executives, politicians, regulators, and labor leaders. version 2 was higher resolution: there were more companies, of more different sizes, making more different things, and their relationships changed faster. in this world there were still plenty of back room negotiations, but more was left to market forces. which further accelerated the fragmentation.

it's a little misleading to talk of versions when describing a gradual process, but not as misleading as it might seem. there was a lot of change in a few decades, and what we ended up with was qualitatively different. the companies in the s&p 500 in 1958 had been there an average of 61 years. by 2012 that number was 18 years. [18]

the breakup of the duplo economy happened simultaneously with the spread of computing power. to what extent were computers a precondition? it would take a book to answer that. obviously the spread of computing power was a precondition for the rise of startups. i suspect it was for most of what happened in finance too. but was it a precondition for globalization or the lbo wave? i don't know, but i wouldn't discount the possibility. it may be that the refragmentation was driven by computers in the way the industrial revolution was driven by steam engines. whether or not computers were a precondition, they have certainly accelerated it.

the new fluidity of companies changed people's relationships with their employers. why climb a corporate ladder that might be yanked out from under you? ambitious people started to think of a career less as climbing a single ladder than as a series of jobs that might be at different companies. more movement (or even potential movement) between companies introduced more competition in salaries. plus as companies became smaller it became easier to estimate how much an employee contributed to the company's revenue. both changes drove salaries toward market price. and since people vary dramatically in productivity, paying market price meant salaries started to diverge.

by no coincidence it was in the early 1980s that the term "yuppie" was coined. that word is not much used now, because the phenomenon it describes is so taken for granted, but at the time it was a label for something novel. yuppies were young professionals who made lots of money. to someone in their twenties today, this wouldn't seem worth naming. why wouldn't young professionals make lots of money? but until the 1980s, being underpaid early in your career was part of what it meant to be a professional. young professionals were paying their dues, working their way up the ladder. the rewards would come later. what was novel about yuppies was that they wanted market price for the work they were doing now.

the first yuppies did not work for startups. that was still in the future. nor did they work for big companies. they were professionals working in fields like law, finance, and consulting. but their example rapidly inspired their peers. once they saw that new bmw 325i, they wanted one too.

underpaying people at the beginning of their career only works if everyone does it. once some employer breaks ranks, everyone else has to, or they can't get good people. and once started this process spreads through the whole economy, because at the beginnings of people's careers they can easily switch not merely employers but industries.

but not all young professionals benefitted. you had to produce to get paid a lot. it was no coincidence that the first yuppies worked in fields where it was easy to measure that.

more generally, an idea was returning whose name sounds old-fashioned precisely because it was so rare for so long: that you could make your fortune. as in the past there were multiple ways to do it. some made their fortunes by creating wealth, and others by playing zero-sum games. but once it bec-

technology is a lever. it magnifies work. and the lever not only grows increasingly long, but the rate at which it grows is itself increasing.

which in turn means the variation in the amount of wealth people can create has not only been increasing, but accelerating. the unusual conditions that prevailed in the mid 20th century masked this underlying trend. the ambitious had little choice but to join large organizations that made them march in step with lots of other people — literally in the case of the armed forces, figuratively in the case of big corporations. even if the big corporations had wanted to pay people proportionate to their value, they couldn't have figured out how. but that constraint has gone now. ever since it started to erode in the 1970s, we've seen the underlying forces at work again. [22]

not everyone who gets rich now does it by creating wealth, certainly. but a significant number do, and the baumol effect means all their peers get dragged along too. [23] and as long as it's possible to get rich by creating wealth, the default tendency will be for economic inequality to increase. even if you eliminate all the other ways to get rich. you can mitigate this with subsidies at the bottom and taxes at the top, but unless taxes are high enough to discourage people from creating wealth, you're always going to be fighting a losing battle against increasing variation in productivity. [24]

that form of fragmentation, like the others, is here to stay. or rather, back to stay. nothing is forever, but the tendency toward fragmentation should be more forever than most things, precisely because it's not due to any particular cause. it's simply a reversion to the mean. when rockefeller said individualism was gone, he was right for a hundred years. it's back now, and that's likely to be true for longer.

i worry that if we don't acknowledge this, we're headed for trouble. if we think 20th century cohesion disappeared because of few policy tweaks, we'll be deluded into thinking we can get it back (minus the bad parts, somehow) with a few countertweaks. and then we'll waste our time trying to eliminate fragmentation, when we'd be better off thinking about how to mitigate its consequences.

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notes

1 lester thurow, writing in 1975, said the wage differentials prevailing at the end of world war ii had become so embedded that they "were regarded as 'just' even after the egalitarian pressures of world war ii had disappeared. basically, the same differentials exist to this day, thirty years later." but goldin and margo think market forces in the postwar period also helped preserve the wartime compression of wages — specifically increased demand for unskilled workers, and oversupply of educated ones.

(oddly enough, the american custom of having employers pay for health insurance derives from efforts by businesses to circumvent nwlb wage controls in order to attract workers.)

2 as always, tax rates don't tell the whole story. there were lots of exemptions, especially for individuals. and in world war ii the tax codes were so new that the government had little acquired immunity to tax avoidance. if the rich paid high taxes during the war it was more because they wanted to than because they had to.

after the war, federal tax receipts as a percentage of gdp were about the same as they are now. in fact, for the entire period since the war, tax receipts have stayed close to 18% of gdp, despite dramatic changes in tax rates. the lowest point occurred when marginal income tax rates were highest: 14.1% in 1950. looking at the data, it's hard to avoid the conclusion that tax rates have had little effect on what people actually paid.

by mid-century big companies funded 3/4 of their growth from earnings. but the early years of bank dependence, reinforced by the financial controls of world war ii, must have had a big effect on social conventions about executive salaries. so it may be that the lack of movement between companies was as much the effect of low salaries as the cause.

incidentally, the switch in the 1920s to financing growth with retained earnings was one cause of the 1929 crash. the banks now had to find someone else to lend to, so they made more margin loans.

15 even now it's hard to get them to. one of the things i find hardest to get into the heads of would-be startup founders is how important it is to do certain kinds of menial work early in the life of a company. doing things that don't scale is to how henry ford got started as a high-fiber diet is to the traditional peasant's diet: they had no choice but to do the right thing, while we have to make a conscious effort.

16 founders weren't celebrated in the press when i was a kid. "our founder" meant a photograph of a severe-looking man with a walrus mustache and a wing collar who had died decades ago. the thing to be when i was a kid was an *executive*. if you weren't around then it's hard to grasp the cachet that term had. the fancy version of everything was called the "executive" model.

17 the wave of hostile takeovers in the 1980s was enabled by a combination of circumstances: court decisions striking down state anti-takeover laws, starting with the supreme court's 1982 decision in edgar v. mite corp.; the reagan administration's comparatively sympathetic attitude toward takeovers; the depository institutions act of 1982, which allowed banks and savings and loans to buy corporate bonds; a new sec rule issued in 1982 (rule 415) that made it possible to bring corporate bonds to market faster; the creation of the junk bond business by michael milken; a vogue for conglomerates in the preceding period that caused many companies to be combined that never should have been; a decade of inflation that left many public companies trading below the value of their assets; and not least, the increasing complacency of managements.

18 foster, richard. "creative destruction whips through corporate america." innosight, february 2012.

19 ceos of big companies may be overpaid. i don't know enough about big companies to say. but it is certainly not impossible for a ceo to make 200x as much difference to a company's revenues as the average employee. look at what steve jobs did for apple when he came back as ceo. it would have been a good deal for the board to give him 95% of the company. apple's market cap the day steve came back in july 1997 was 1.73 billion. 5% of apple now (january 2016) would be worth about 30 billion. and it would not be if steve hadn't come back; apple probably wouldn't even exist anymore.

merely including steve in the sample might be enough to answer the question of whether public company ceos in the aggregate are overpaid. and that is not as facile a trick as it might seem, because the broader your holdings, the more the aggregate is what you care about.

20 the late 1960s were famous for social upheaval. but that was more rebellion (which can happen in any era if people are provoked sufficiently) than fragmentation. you're not seeing fragmentation unless you see people breaking off to both left and right.

21 globally the trend has been in the other direction. while the us is becoming more fragmented, the world as a whole is becoming less fragmented, and mostly in good ways.

22 there were a handful of ways to make a fortune in the mid 20th century. the main one was drilling for oil, which was open to newcomers because it was not something big companies could dominate through economies of scale. how did individuals accumulate large fortunes in an era of such high taxes? giant tax loopholes defended by two of the most powerful men in congress, sam rayburn and lyndon johnson.

but becoming a texas oilman was not in 1950 something one could aspire to the way starting a startup or going to work on wall street were in 2000, because (a) there was a strong local component and (b) success depended so much on luck.

23 the baumol effect induced by startups is very visible in silicon valley. google will pay people millions of dollars a year to keep them from leaving to start or join startups.

07

cities and ambition

may 2008
paulgraham.com/cities.html

american universities currently seem to be the best, judging from the flow of ambitious students. and what us city has a stronger claim? new york? a fair number of smart people, but diluted by a much larger number of neanderthals in suits. the bay area has a lot of smart people too, but again, diluted; there are two great universities, but they're far apart. harvard and mit are practically adjacent by west coast standards, and they're surrounded by about 20 other colleges and universities. [1]

cambridge as a result feels like a town whose main industry is ideas, while new york's is finance and silicon valley's is startups.

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when you talk about cities in the sense we are, what you're really talking about is collections of people. for a long time cities were the only large collections of people, so you could use the two ideas interchangeably. but we can see how much things are changing from the examples i've mentioned. new york is a classic great city. but cambridge is just part of a city, and silicon valley is not even that. (san jose is not, as it sometimes claims, the capital of silicon valley. it's just 178 square miles at one end of it.)

maybe the internet will change things further. maybe one day the most important community you belong to will be a virtual one, and it won't matter where you live physically. but i wouldn't bet on it. the physical world is very high bandwidth, and some of the ways cities send you messages are quite subtle.

one of the exhilarating things about coming back to cambridge every spring is walking through the streets at dusk, when you can see into the houses. when you walk through palo alto in the evening, you see nothing but the blue glow of tvs. in cambridge you see shelves full of promising-looking books. palo alto was probably much like cambridge in 1960, but you'd never guess now that there was a university nearby. now it's just one of the richer neighborhoods in silicon valley. [2]

a city speaks to you mostly by accident — in things you see through windows, in conversations you overhear. it's not something you have to seek out, but something you can't turn off. one of the occupational hazards of living in cambridge is overhearing the conversations of people who use interrogative intonation in declarative sentences. but on average i'll take cambridge conversations over new york or silicon valley ones.

a friend who moved to silicon valley in the late 90s said the worst thing about living there was the low quality of the eavesdropping. at the time i thought she was being deliberately eccentric. sure, it can be interesting to eavesdrop on people, but is good quality eavesdropping so important that it would affect where you chose to live? now i understand what she meant. the conversations you overhear tell you what sort of people you're among.

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no matter how determined you are, it's hard not to be influenced by the people around you. it's not so much that you do whatever a city expects of you, but that you get discouraged when no one around you cares about the same things you do.

there's an imbalance between encouragement and discouragement like that between gaining and losing money. most people overvalue negative amounts of money: they'll work much harder to avoid losing a dollar than to gain one. similarly, although there are plenty of people strong enough to resist doing something just because that's what one is supposed to do where they happen to be, there are few strong enough to keep working on something no one around them cares about.

because ambitions are to some extent incompatible and admiration is a zero-sum game, each city tends to focus on one type of ambition. the reason cambridge is the intellectual capital is not just that there's

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so far the complete list of messages i've picked up from cities is: wealth, style, hipness, physical attractiveness, fame, political power, economic power, intelligence, social class, and quality of life.

my immediate reaction to this list is that it makes me slightly queasy. i'd always considered ambition a good thing, but i realize now that was because i'd always implicitly understood it to mean ambition in the areas i cared about. when you list everything ambitious people are ambitious about, it's not so pretty.

on closer examination i see a couple things on the list that are surprising in the light of history. for example, physical attractiveness wouldn't have been there 100 years ago (though it might have been 2400 years ago). it has always mattered for women, but in the late twentieth century it seems to have started to matter for men as well. i'm not sure why — probably some combination of the increasing power of women, the increasing influence of actors as models, and the fact that so many people work in offices now: you can't show off by wearing clothes too fancy to wear in a factory, so you have to show off with your body instead.

hipness is another thing you wouldn't have seen on the list 100 years ago. or wouldn't you? what it means is to know what's what. so maybe it has simply replaced the component of social class that consisted of being "au fait." that could explain why hipness seems particularly admired in london: it's version 2 of the traditional english delight in obscure codes that only insiders understand.

economic power would have been on the list 100 years ago, but what we mean by it is changing. it used to mean the control of vast human and material resources. but increasingly it means the ability to direct the course of technology, and some of the people in a position to do that are not even rich — leaders of important open source projects, for example. the captains of industry of times past had laboratories full of clever people cooking up new technologies for them. the new breed are themselves those people.

as this force gets more attention, another is dropping off the list: social class. i think the two changes are related. economic power, wealth, and social class are just names for the same thing at different stages in its life: economic power converts to wealth, and wealth to social class. so the focus of admiration is simply shifting upstream.

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does anyone who wants to do great work have to live in a great city? no; all great cities inspire some sort of ambition, but they aren't the only places that do. for some kinds of work, all you need is a handful of talented colleagues.

what cities provide is an audience, and a funnel for peers. these aren't so critical in something like math or physics, where no audience matters except your peers, and judging ability is sufficiently straightforward that hiring and admissions committees can do it reliably. in a field like math or physics all you need is a department with the right colleagues in it. it could be anywhere — in los alamos, new mexico, for example.

it's in fields like the arts or writing or technology that the larger environment matters. in these the best practitioners aren't conveniently collected in a few top university departments and research labs — partly because talent is harder to judge, and partly because people pay for these things, so one doesn't need to rely on teaching or research funding to support oneself. it's in these more chaotic fields that it helps most to be in a great city: you need the encouragement of feeling that people around you care

chicago, 8 in london, 13 in new york, and 20 in paris.

(zagat's lists the ritz carlton dining room in sf as requiring jackets but i couldn't believe it, so i called to check and in fact they don't. apparently there's only one restaurant left on the entire west coast that still requires jackets: the french laundry in napa valley.)

4 ideas are one step upstream from economic power, so it's conceivable that intellectual centers like cambridge will one day have an edge over silicon valley like the one the valley has over new york.

this seems unlikely at the moment; if anything boston is falling further and further behind. the only reason i even mention the possibility is that the path from ideas to startups has recently been getting smoother. it's a lot easier now for a couple of hackers with no business experience to start a startup than it was 10 years ago. if you extrapolate another 20 years, maybe the balance of power will start to shift back. i wouldn't bet on it, but i wouldn't bet against it either.

5 if paris is where people care most about art, why is new york the center of gravity of the art business? because in the twentieth century, art as brand split apart from art as stuff. new york is where the richest buyers are, but all they demand from art is brand, and since you can base brand on anything with a sufficiently identifiable style, you may as well use the local stuff.

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08

the four quadrants of conformism

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paulgraham.com/conformism.html

he's too polite to say so, but of course they wouldn't. and indeed, our default assumption should not merely be that his students would, on average, have behaved the same way people did at the time, but that the ones who are aggressively conventional-minded today would have been aggressively conventional-minded then too. in other words, that they'd not only not have fought against slavery, but that they'd have been among its staunchest defenders.

i'm biased, i admit, but it seems to me that aggressively conventional-minded people are responsible for a disproportionate amount of the trouble in the world, and that a lot of the customs we've evolved since the enlightenment have been designed to protect the rest of us from them. in particular, the retirement of the concept of heresy and its replacement by the principle of freely debating all sorts of different ideas, even ones that are currently considered unacceptable, without any punishment for those who try them out to see if they work. [2]

why do the independent-minded need to be protected, though? because they have all the new ideas. to be a successful scientist, for example, it's not enough just to be right. you have to be right when everyone else is wrong. conventional-minded people can't do that. for similar reasons, all successful startup ceos are not merely independent-minded, but aggressively so. so it's no coincidence that societies prosper only to the extent that they have customs for keeping the conventional-minded at bay. [3]

in the last few years, many of us have noticed that the customs protecting free inquiry have been weakened. some say we're overreacting — that they haven't been weakened very much, or that they've been weakened in the service of a greater good. the latter i'll dispose of immediately. when the conventional-minded get the upper hand, they always say it's in the service of a greater good. it just happens to be a different, incompatible greater good each time.

as for the former worry, that the independent-minded are being oversensitive, and that free inquiry hasn't been shut down that much, you can't judge that unless you are yourself independent-minded. you can't know how much of the space of ideas is being lopped off unless you have them, and only the independent-minded have the ones at the edges. precisely because of this, they tend to be very sensitive to changes in how freely one can explore ideas. they're the canaries in this coalmine.

the conventional-minded say, as they always do, that they don't want to shut down the discussion of all ideas, just the bad ones.

you'd think it would be obvious just from that sentence what a dangerous game they're playing. but i'll spell it out. there are two reasons why we need to be able to discuss even "bad" ideas.

the first is that any process for deciding which ideas to ban is bound to make mistakes. all the more so because no one intelligent wants to undertake that kind of work, so it ends up being done by the stupid. and when a process makes a lot of mistakes, you need to leave a margin for error. which in this case means you need to ban fewer ideas than you'd like to. but that's hard for the aggressively conventional-minded to do, partly because they enjoy seeing people punished, as they have since they were children, and partly because they compete with one another. enforcers of orthodoxy can't allow a borderline idea to exist, because that gives other enforcers an opportunity to one-up them in the moral purity department, and perhaps even to turn enforcer upon them. so instead of getting the margin for error we need, we get the opposite: a race to the bottom in which any idea that seems at all bannable ends up being banned. [4]

the second reason it's dangerous to ban the discussion of ideas is that ideas are more closely related than they look. which means if you restrict the discussion of some topics, it doesn't only affect those topics. the restrictions propagate back into any topic that yields implications in the forbidden ones. and

4 there has been progress in one area: the punishments for talking about banned ideas are less severe than in the past. there's little danger of being killed, at least in richer countries. the aggressively conventional-minded are mostly satisfied with getting people fired.

5 many professors are independent-minded — especially in math, the hard sciences, and engineering, where you have to be to succeed. but students are more representative of the general population, and thus mostly conventional-minded. so when professors and students are in conflict, it's not just a conflict between generations but also between different types of people.

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09

orthodox privilege

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10

the two kinds of moderate

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ident, or more resigned, or simply more disorganized. i just don't know.

3 having heretical opinions doesn't mean you have to express them openly. it may be easier to have them if you don't.

thanks to austen allred, trevor blackwell, patrick collison, jessica livingston, amjad masad, ryan petersen, and harj taggar for reading drafts of this.

11

haters

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talent is what drives some people to become haters. they're not just saying "it's unfair that so-and-so is famous," but "it's unfair that so-and-so is famous, and not me."

could a hater be cured if they achieved something impressive? my guess is that's a moot point, because they never will. i've been able to observe for long enough that i'm fairly confident the pattern works both ways: not only do people who do great work never become haters, haters never do great work. although i dislike the word "fanboy," it's evocative of something important about both haters and fanboys. it implies that the fanboy is so slavishly predictable in his admiration that he's diminished as a result, that he's less than a man.

haters seem even more diminished. i can imagine being a fanboy. i can think of people whose work i admire so much that i could abase myself before them out of sheer gratitude. if p. g. wodehouse were still alive, i could see myself being a wodehouse fanboy. but i could not imagine being a hater.

knowing that haters are just fanboys with the sign bit flipped makes it much easier to deal with them. we don't need a separate theory of haters. we can just use existing techniques for dealing with obsessive fans.

the most important of which is simply not to think much about them. if you're like most people who become famous enough to acquire haters, your initial reaction will be one of mystification. why does this guy seem to have it in for me? where does his obsessive energy come from, and what makes him so appallingly nasty? what did i do to set him off? is it something i can fix?

the mistake here is to think of the hater as someone you have a dispute with. when you have a dispute with someone, it's usually a good idea to try to understand why they're upset and then fix things if you can. disputes are distracting. but it's a false analogy to think of a hater as someone you have a dispute with. it's an understandable mistake, if you've never encountered haters before. but when you realize that you're dealing with a hater, and what a hater is, it's clear that it's a waste of time even to think about them. if you have obsessive fans, do you spend any time wondering what makes them love you so much? no, you just think "some people are kind of crazy," and that's the end of it.

since haters are equivalent to fanboys, that's the way to deal with them too. there may have been something that set them off. but it's not something that would have set off a normal person, so there's no reason to spend any time thinking about it. it's not you, it's them.

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notes

1 there are of course some people who are genuine frauds. how can you distinguish between x calling y a fraud because x is a hater, and because y is a fraud? look at neutral opinion. actual frauds are usually pretty conspicuous. thoughtful people are rarely taken in by them. so if there are some thoughtful people who like y, you can usually assume y is not a fraud.

2 i would make an exception for teenagers, who sometimes act in such extreme ways that they are literally not themselves. i can imagine a teenage kid being a hater and then growing out of it. but not anyone over 25.

3 i have a much worse memory for misdeeds than my wife jessica, who is a connoisseur of character, but i don't wish it were better. most disputes are a waste of time even if you're in the right, and it's easy to bury the hatchet with someone if you can't remember why you were mad at them.

4 a competent hater will not merely attack you individually but will try to get mobs after you. in some cases you may want to refute whatever bogus claim they made in order to do so. but err on the side of not, because

12

the origins of wokeness

january 2025
paulgraham.com/woke.html

them and the previous generation of professors started to retire, they gradually became both.

the reason political correctness began in the humanities and social sciences was that these fields offered more scope for the injection of politics. a 1960s radical who got a job as a physics professor could still attend protests, but his political beliefs wouldn't affect his work. whereas research in sociology and modern literature can be made as political as you like. [2]

i saw political correctness arise. when i started college in 1982 it was not yet a thing. female students might object if someone said something they considered sexist, but no one was getting *reported* for it. it was still not a thing when i started grad school in 1986. it was definitely a thing in 1988 though, and by the early 1990s it seemed to pervade campus life.

what happened? how did protest become punishment? why were the late 1980s the point at which protests against male chauvinism (as it used to be called) morphed into formal complaints to university authorities about sexism? basically, the 1960s radicals got tenure. they became the establishment they'd protested against two decades before. now they were in a position not just to speak out about their ideas, but to enforce them.

a new set of moral rules to enforce was exciting news to a certain kind of student. what made it particularly exciting was that they were allowed to attack professors. i remember noticing that aspect of political correctness at the time. it wasn't simply a grass-roots student movement. it was faculty members encouraging students to attack other faculty members. in that respect it was like the cultural revolution. that wasn't a grass-roots movement either; that was mao unleashing the younger generation on his political opponents. and in fact when roderick macfarquhar started teaching a class on the cultural revolution at harvard in the late 1980s, many saw it as a comment on current events. i don't know if it actually was, but people thought it was, and that means the similarities were obvious. [3]

college students larp. it's their nature. it's usually harmless. but larping morality turned out to be a poisonous combination. the result was a kind of moral etiquette, superficial but very complicated. imagine having to explain to a well-meaning visitor from another planet why using the phrase "people of color" is considered particularly enlightened, but saying "colored people" gets you fired. and why exactly one isn't supposed to use the word "negro" now, even though martin luther king used it constantly in his speeches. there are no underlying principles. you'd just have to give him a long list of rules to memorize. [4]

the danger of these rules was not just that they created land mines for the unwary, but that their elaborateness made them an effective substitute for virtue. whenever a society has a concept of heresy and orthodoxy, orthodoxy becomes a substitute for virtue. you can be the worst person in the world, but as long as you're orthodox you're better than everyone who isn't. this makes orthodoxy very attractive to bad people.

but for it to work as a substitute for virtue, orthodoxy must be difficult. if all you have to do to be orthodox is wear some garment or avoid saying some word, everyone knows to do it, and the only way to seem more virtuous than other people is to actually be virtuous. the shallow, complicated, and frequently changing rules of political correctness made it the perfect substitute for actual virtue. and the result was a world in which good people who weren't up to date on current moral fashions were brought down by people whose characters would make you recoil in horror if you could see them.

one big contributing factor in the rise of political correctness was the lack of other things to be morally pure about. previous generations of prigs had been prigs mostly about religion and sex. but among the cultural elite these were the deadest of dead letters by the 1980s; if you were religious, or a virgin, this

world, although it still burned hottest within universities. and it was concerned with a wider variety of sins. in the first phase of political correctness there were really only three things people got accused of: sexism, racism, and homophobia (which at the time was a neologism invented for the purpose). but between then and 2010 a lot of people had spent a lot of time trying to invent new kinds of -isms and -phobias and seeing which could be made to stick.

the second phase was, in multiple senses, political correctness metastasized. why did it happen when it did? my guess is that it was due to the rise of social media, particularly tumblr and twitter, because one of the most distinctive features of the second wave of political correctness was the *cancel mob*: a mob of angry people uniting on social media to get someone ostracized or fired. indeed this second wave of political correctness was originally called "cancel culture"; it didn't start to be called "wokeness" till the 2020s.

one aspect of social media that surprised almost everyone at first was the popularity of outrage. users seemed to *like* being outraged. we're so used to this idea now that we take it for granted, but really it's pretty strange. being outraged is not a pleasant feeling. you wouldn't expect people to seek it out. but they do. and above all, they want to share it. i happened to be running a forum from 2007 to 2014, so i can actually quantify how much they want to share it: our users were about three times more likely to upvote something if it outraged them.

this tilt toward outrage wasn't due to wokeness. it's an inherent feature of social media, or at least this generation of it. but it did make social media the perfect mechanism for fanning the flames of wokeness. [7]

it wasn't just public social networks that drove the rise of wokeness though. group chat apps were also critical, especially in the final step, cancellation. imagine if a group of employees trying to get someone fired had to do it using only email. it would be hard to organize a mob. but once you have group chat, mobs form naturally.

another contributing factor in this second wave of political correctness was the dramatic increase in the polarization of the press. in the print era, newspapers were constrained to be, or at least seem, politically neutral. the department stores that ran ads in the new york times wanted to reach everyone in the region, both liberal and conservative, so the times had to serve both. but the times didn't regard this neutrality as something forced upon them. they embraced it as their duty as a *paper of record* — as one of the big newspapers that aimed to be chronicles of their times, reporting every sufficiently important story from a neutral point of view.

when i grew up the papers of record seemed timeless, almost sacred institutions. papers like the new york times and washington post had immense prestige, partly because other sources of news were limited, but also because they did make some effort to be neutral.

unfortunately it turned out that the paper of record was mostly an artifact of the constraints imposed by print. [8] when your market was determined by geography, you had to be neutral. but publishing online enabled — in fact probably forced — newspapers to switch to serving markets defined by ideology instead of geography. most that remained in business fell in the direction they'd already been leaning: left. on october 11, 2020 the new york times announced that "the paper is in the midst of an evolution from the stodgy paper of record into a juicy collection of great narratives." [9] meanwhile journalists, of a sort, had arisen to serve the right as well. and so journalism, which in the previous era had been one of the great centralizing forces, now became one of the great polarizing ones.

a rule, their default reaction is to believe you. especially if multiple people tell them. which in turn is a recipe for exponential growth. zealots invent some new impropriety to avoid. the first people to adopt it are fellow zealots, eager for new ways to signal their virtue. if there are enough of these, the initial group of zealots is followed by a much larger group, motivated by fear. they're not trying to signal virtue; they're just trying to avoid getting in trouble. at this point the new impropriety is now firmly established. plus its success has increased the rate of change in social rules, which, remember, is one of the reasons people are nervous about which rules they might be breaking. so the cycle accelerates. [12]

what's true of individuals is even more true of organizations. especially organizations without a powerful leader. such organizations do everything based on "best practices." there's no higher authority; if some new "best practice" achieves critical mass, they *must* adopt it. and in this case the organization can't do what it usually does when it's uncertain: delay. it might be committing improprieties right now! so it's surprisingly easy for a small group of zealots to capture this type of organization by describing new improprieties it might be guilty of. [13]

how does this kind of cycle ever end? eventually it leads to disaster, and people start to say enough is enough. the excesses of 2020 made a lot of people say that.

since then wokeness has been in gradual but continual retreat. corporate ceos, starting with brian armstrong, have openly rejected it. universities, led by the university of chicago and mit, have explicitly confirmed their commitment to free speech. twitter, which was arguably the hub of wokeness, was bought by elon musk in order to neutralize it, and he seems to have succeeded — and not, incidentally, by censoring left-wing users the way twitter used to censor right-wing ones, but without censoring either. [14] consumers have emphatically rejected brands that ventured too far into wokeness. the bud light brand may have been permanently damaged by it. i'm not going to claim trump's second victory in 2024 was a referendum on wokeness; i think he won, as presidential candidates always do, because he was more charismatic; but voters' disgust with wokeness must have helped.

so what do we do now? wokeness is already in retreat. obviously we should help it along. what's the best way to do that? and more importantly, how do we avoid a third outbreak? after all, it seemed to be dead once, but came back worse than ever.

in fact there's an even more ambitious goal: is there a way to prevent any similar outbreak of aggressively performative moralism in the future — not just a third outbreak of political correctness, but the next thing like it? because there will be a next thing. prigs are prigs by nature. they need rules to obey and enforce, and now that darwin has cut off their traditional supply of rules, they're constantly hungry for new ones. all they need is someone to meet them halfway by defining a new way to be morally pure, and we'll see the same phenomenon again.

let's start with the easier problem. is there a simple, principled way to deal with wokeness? i think there is: to use the customs we already have for dealing with religion. wokeness is effectively a religion, just with god replaced by protected classes. it's not even the first religion of this kind; marxism had a similar form, with god replaced by the masses. [15] and we already have well-established customs for dealing with religion within organizations. you can express your own religious identity and explain your beliefs, but you can't call your coworkers infidels if they disagree, or try to ban them from saying things that contradict its doctrines, or insist that the organization adopt yours as its official religion.

if we're not sure what to do about any particular manifestation of wokeness, imagine we were dealing with some other religion, like christianity. should we have people within organizations whose jobs are to enforce woke orthodoxy? no, because we wouldn't have people whose jobs were to enforce christian

fight back at that point, with any luck we can stop them in their tracks.

the number of true things we can't say should not increase. if it does, something is wrong.

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notes

0 this was not the original meaning of "woke," but it's rarely used in the original sense now. now the pejorative sense is the dominant one.

1 why did 1960s radicals focus on the causes they did? one of the people who reviewed drafts of this essay explained this so well that i asked if i could quote him:

> the middle-class student protestors of the new left rejected the socialist/marxist left as unhip. they were interested in sexier forms of oppression uncovered by cultural analysis (marcuse) and abstruse "theory". labor politics became stodgy and old-fashioned. this took a couple generations to work through. the woke ideology's conspicuous lack of interest in the working class is the tell-tale sign. such fragments as are, er, left of the old left are anti-woke, and meanwhile the actual working class shifted to the populist right and gave us trump. trump and wokeness are cousins.
>
> the middle-class origins of wokeness smoothed its way through the institutions because it had no interest in "seizing the means of production" (how quaint such phrases seem now), which would quickly have run up against hard state and corporate power. the fact that wokeness only expressed interest in other kinds of class (race, sex, etc) signalled compromise with existing power: give us power within your system and we'll bestow the resource we control — moral rectitude — upon you. as an ideological stalking horse for gaining control over discourse and institutions, this succeeded where a more ambitious revolutionary program would not have.

2 it helped that the humanities and social sciences also included some of the biggest and easiest undergrad majors. if a political movement had to start with physics students, it could never get off the ground; there would be too few of them, and they wouldn't have the time to spare.

at the top universities these majors are not as big as they used to be, though. a 2022 survey found that only 7% of harvard undergrads plan to major in the humanities, vs nearly 30% during the 1970s. i expect wokeness is at least part of the reason; when undergrads consider majoring in english, it's presumably because they love the written word and not because they want to listen to lectures about racism.

3 the puppet-master-and-puppet character of political correctness became clearly visible when a bakery near oberlin college was falsely accused of race discrimination in 2016. in the subsequent civil trial, lawyers for the bakery produced a text message from oberlin dean of students meredith raimondo that read "i'd say unleash the students if i wasn't convinced this needs to be put behind us."

4 the woke sometimes claim that wokeness is simply treating people with respect. but if it were, that would be the only rule you'd have to remember, and this is comically far from being the case. my younger son likes to imitate voices, and at one point when he was about seven i had to explain which accents it was currently safe to imitate publicly and which not. it took about ten minutes, and i still hadn't covered all the cases.

5 in 1986 the supreme court ruled that creating a hostile work environment could constitute sex discrimination, which in turn affected universities via title ix. the court specified that the test of a hostile environment was whether it would bother a reasonable person, but since for a professor merely being the subject of a sexual harassment complaint would be a disaster whether the complainant was reasonable or not, in practice any joke or remark remotely connected with sex was now effectively forbidden. which meant we'd now come full circle to victorian codes of behavior, when there was a large class of things that might not be said "with ladies present."

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13

the brand age

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paulgraham.com/brandage.html

present-day watchmakers all know but rarely come so close to saying outright: whatever age we're in now, it's not the golden age.

the golden age was from 1945 to 1970 — from the point where the watch industry emerged from the chaos of war with the swiss on top till the triple cataclysm that struck it starting in the late 60s. there were two things watchmakers sought above all in the golden age: thinness and accuracy. and indeed this was arguably the essential tradeoff in watchmaking. a watch is something you carry with you to tell you the time. so there are two fundamental ways to improve it: to make it easier to carry with you and to make it better at telling the time.

obviously accuracy is valuable, but in the golden age thinness was if anything more valuable. even in the days of pocket watches the best watchmakers tried to make their watches as thin as they could. cheap, thick pocket watches were derided as "turnips." but thinness took on a new urgency when men's watches moved onto their wrists during world war i. and since thinness was more difficult to achieve than accuracy, it was this quality that tended to distinguish the more expensive watches of the golden age.

there is one other thing watchmakers have pursued in some eras: telling more than the time in the usual way. telling you the phase of the moon, for example, or telling the time with sound. in the industry the term for these things is "complications." they were popular in the nineteenth century and they're popular again now, but except for one pragmatic complication (showing the date), they were a sideshow in the golden age. in the golden age, as always in golden ages, the top watchmakers focused on the essential tradeoff. and, as always in golden ages, they did it beautifully. the best watches of the golden age have a quiet perfection that has never been equalled since. and for reasons i'm about to explain, probably never will be.

the three most prestigious brands of the golden age were the so-called "holy trinity" of patek philippe, vacheron constantin, and audemars piguet. their prestige was mostly deserved; they had earned it by the exceptional quality of their work. by the 1960s they stood on two legs, prestige and performance. and what they learned in the next two decades was that they had to put all their weight on the first leg, because they could no longer win at either of the two things watchmakers had historically striven to achieve. quartz movements were not only more accurate than any mechanical movement, but thinner too.

the holy trinity at least had another leg to stand on. most of the other well-known swiss watchmakers sold only performance. none of those companies survived intact.

omega showed what not to do. omega were the nerds of swiss watchmakers. they made wonderfully accurate watches, but they would have been ambivalent, at best, about the idea of being a luxury brand. when the japanese got as good as the swiss at making accurate movements, omega responded in the omega way: make even more accurate movements. they introduced a new movement in 1968 that ran at a 45% higher frequency. in theory this should have made it more accurate, but the new movement was so fragile that it destroyed their reputation for reliability. they even tried to make a better quartz movement, but there was nothing down that road but a race to the bottom. by 1981 they were insolvent and were taken over by their creditors.

patek philippe took the opposite approach. while omega was redesigning their movements, patek was redesigning their cases. or more precisely, designing their cases, because until then they hadn't.

this is probably the point to mention what a strange beast the swiss watch industry was in those days. it was a kind of capitalism that's hard to imagine today, and even then could only have been made to

indeed, the conflict between branding and design is so fundamental that it extends far beyond things we call design. we see it even in religion. if you want the adherents of a religion to have customs that set them apart from everyone else, you can't make them do things that are convenient or reasonable, or other people would do them too. if you want to set your adherents apart, you have to make them do things that are inconvenient and unreasonable.

it's the same if you want to set your designs apart. if you choose good options, other people will choose them too.

there are only two ways to combine branding and good design. you can do it when the space of possibilities is enormously large, as it is in painting, for example. leonardo could paint as well as he possibly could and yet also paint in a style that was distinctively his. if there had been a million painters as good as bellini and leonardo this would have been harder to do, but since there were more like ten they didn't bump up against one another much. [4]

the other situation when branding and good design can be combined is when the space of possibilities is comparatively unexplored. if you're the first to arrive in some new territory, you can both find the right answer and claim it as uniquely yours. at least at first; if you've really found the right answer, everyone else's designs will inevitably converge on yours, and your brand advantage will erode over time.

since the space of watch design is neither unexplored nor enormously large, branding can only be achieved at the expense of good design. and in fact if you wanted one sentence to describe the current age of watchmaking, that one would do pretty well.

patek philippe didn't know for sure that making visibly branded watches would work. it was not even their only strategy, at the time. they were finding their way. but it was the strategy that did work, at least as measured by revenues.

for it to work the customers had to meet them halfway. patek knew that not all their customers were buying their watches for the performance they delivered — for their accuracy and thinness. they knew that at least some customers were buying them because they were expensive. but it was unclear how many, or how far they could be pushed.

to encourage them, patek did something that none of the holy trinity had done much of before: brand advertising. and what they talked about was how expensive their watches were. a 1968 patek ad explained "why you are well advised to invest perhaps half a month's income" in an ellipse. "like every patek philippe," the ad continued, "this thin model is entirely finished by hand. since a patek philippe is the costliest watch to make, production is severely limited: only 43 watches are signed out each day for delivery to prominent jewelers throughout the world." [5]

you can tell this is an early ad because they still mention thinness. but there is no mention of accuracy. presumably patek felt that battle was already lost.

the next move was made by audemars piguet, who in 1970 commissioned the renowned designer gérald genta to design their own iconic watch, this one, daringly, in steel. the result, launched in 1972, was the royal oak. and audemars piguet's ads (for they too now started doing brand advertising) emphasized its high cost even more dramatically. "introducing steel at the price of gold," one began. "you're looking at the costliest stainless steel watch in the world — the audemars piguet 'royal oak'. what makes it even more precious than gold is the time that went into building it, by a vanishing breed of master watchmakers." at the bottom of the ad they turn the traditional formula on its head and describe their watches as being "priced from $35,000 and down."

mechanical watches. by 1990 it was. by 1990 the custom of using expensive, highly-branded, conspicuously mechanical watches as status symbols was firmly established. [7]

obsolete technologies don't usually get adopted as ways to display wealth. why did it happen with mechanical watches? because the wristwatch turns out to be the perfect vehicle for it. where better than right on your wrist, where everyone can see it? and more to the point, what better to do it with? you could wear a diamond ring or a gold chain, but those would have seemed socially dubious to investment bankers. they might have been barbarians, but they weren't mafia. whereas nothing could be more legit than a gold watch. the chairman of the company was still wearing one his wife gave him 20 years ago, before quartz watches were even a thing. if the increasing pressure to display wealth was going to emerge anywhere, this was the place. [8]

for men, at least. women never really went for the idea of wearing mechanical watches. most rich women are happy wearing a cartier tank with a quartz movement. why the difference? partly for the same reason that most buyers of steam engines are men. but the main reason is that expensive mechanical watches now serve as de facto jewelry for men, and women don't need de facto jewelry because they can wear actual jewelry.

it was critical, though, that mechanical watches were accurate *enough*. a new 3919 would have been off by no more than 5 seconds a day. that was nowhere near as good as quartz. even the cheapest mass market quartz watches were accurate to half a second a day, and the best ones were accurate to 3 seconds a year. but in practice you didn't need that kind of accuracy. if mechanical watches had only been accurate to a minute a day they couldn't have made the leap from keeping time to displaying wealth. it would have seemed too manifestly unluxurious to have a watch that always had the wrong time. but 5 seconds a day was close enough. [9]

this is an important point about the relationship between brand and quality. quality doesn't stop mattering when a product switches to something people buy for its brand. but the way it matters changes shape. it becomes a threshold. it no longer has to be so great that it sells the product; brand sells the product; but it does have to be good enough to maintain the brand's reputation. the brand must not break character.

it was a lucky thing for the watchmakers that yuppies arose just in time to save them. or maybe not so lucky. because the evolution of the market that yuppies represented has continued with a vengeance, and watchmakers have perforce been dragged along with it. if they don't make gigantic blingy watches for buyers in hong kong and dubai, someone else will. so that is what they now find themselves doing. and what began with a few comparatively subtle examples of the conflict between branding and design is now an all out war on design.

the present era of mechanical watchmaking doesn't yet have a name. but if we need one, it's obvious what it should be: the brand age. the golden age ran from 1945 to 1970, followed by the quartz crisis from 1970 to 1985. since 1985 we've been in the brand age.

this won't be the only brand age. indeed, it's not even the first; fine art has been in its own brand age since the establishment of the barr canon in the 1930s. and since we'll probably see more of this kind of thing, it would be worth taking some time to look at what a brand age is like.

how are things different now from the way they were in the golden age? the best way to answer that might be to imagine what someone from the golden age would notice if we brought him here in a time machine.

and they were far ahead of other watchmakers in that department. they already had in the 1940s what we saw patek philippe and audemars piguet struggling to create in the 1970s and 80s: a case that immediately proclaimed the brand of the maker. the rolex look seems to have evolved organically, but once it did, they realized how important it was. in fact they pitched it as one of the features of their watches. a 1960s rolex ad says "you can recognize its classic shape, carved out of a block of solid gold, from the other end of the conference table."

indeed rolex was ahead of its time in both dimensions: their cases were not merely recognizable, but big too, at least by golden age standards. that was not the result of clever marketing, though. it was a byproduct of the founder hans wilsdorf's obsession with building waterproof watches.

as its name suggests, that was the raison d'etre of the rolex oyster. watches like the oyster were designed to be tough, like jeeps. in the golden age there were two poles of watch design. at one end were tool watches, which were thick, tough, and usually made of steel. at the other end were dress watches, which were thin, elegant, and usually made of gold. but rolex blurred the line between them. when they made thick, tough watches, they made them out of gold as well as steel. the result was a sort of luxury jeep. and if that phrase didn't ring a bell in your head, stop and think about it, because that is exactly what everyone is driving now. that's what suvs are, luxury jeeps. what happened to watches is the same thing that happened to cars. and indeed if our time traveler turned and saw a porsche cayenne pass by and realized what it was — a huge, pseudo-offroad vehicle meant to recall the porsche 911 — he might have been even more shocked than he was by the watches he'd been looking at. [14]

if the time traveller walked into a patek philippe boutique and actually tried to buy a nautilus, he'd get the biggest shock of all. they wouldn't sell him one. because at patek he'd encounter the most extreme brand age phenomenon: artificial scarcity. you can't just buy a nautilus. you have to spend years proving your loyalty first by buying your way through multiple tiers of other models, and then spend years on a waiting list. [15]

obviously this strategy sells more watches. but it also supports retail prices by keeping watches off the secondary market. a company using artificial scarcity to drive sales can't allow too many of the scarce models to leak into the secondary market, or they stop being scarce. the ideal is the watch equivalent of carbon sequestration: for the people who buy their watches to keep them till they die.

to push the market toward this ideal, patek squeezes from both sides of the sale. they weed out flippers by making the path to the scarce models so costly in both time and money — so inconvenient and unreasonable — that only a genuine fan would endure it. the lower tier watches sell for below retail on the secondary market, because patek *doesn't* restrict their supply, so a would-be flipper should have to spend years making money-losing purchases before he could even get something he could flip at a profit. apparently some people still manage to beat this system though, so patek's countermeasures don't end there. they keep a vigilant eye on secondary sales to see who's selling their watches. auction listings usually include serial numbers, so those are easy to trace, but if necessary they'll rebuy their own watches on the secondary market to get the serial number and trace the leak. they buy hundreds a year. and when they catch someone selling watches they don't want them to, they don't just cut off that customer. if a retailer's customers are responsible for too many such leaks, they'll cut off the whole retailer. which naturally makes retailers eager to help them police buyers.

there will of course always be some leaks into the secondary market. even the most loyal customers die at a certain rate. and in fact it's critical for patek that the secondary market continue to exist, because it's one of the most valuable sources of information they have about the most important question they face: how fast to increase the supply of the top tier watches. their scarcity helps drive the purchases of

in fact there's a single principle that will both save you from working on things like brand, and also automatically find golden ages for you. follow the problems.

the way to find golden ages is not to go looking for them. the way to find them — the way almost all their participants have found them historically — is by following interesting problems. if you're smart and ambitious and honest with yourself, there's no better guide than your taste in problems. go where interesting problems are, and you'll probably find that other smart and ambitious people have turned up there too. and later they'll look back on what you did together and call it a golden age.

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notes

1 the bretton woods agreement didn't fix exchange rates between currencies directly. it fixed each relative to gold. obviously this also fixed them relative to one another.

2 the golden ellipse isn't quite a round rect, because the sides aren't quite flat. it's similar in shape to the superellipses popularized by piet hein in the early 1960s, and in fact that may be where they got the name. but mathematically it's not an actual superellipse. my guess is that patek's designer just experimented with french curves till he got something he liked. and to be fair it is a good shape.

3 it was ironic that patek philippe of all companies made this mistake, because adrien philippe was the inventor of the modern crown. but they must have realized what they'd done, because later ellipses have if anything excessively prominent crowns.

4 the high ratio of design space to practitioners in fine art has combined with the practical importance of attribution to give people the impression that painting in a distinctively leonardesque way is what makes leonardo good. the most dangerous problem faced by curators, art historians, and art dealers — the one that has the worst consequences if they get the wrong answer — is attribution. so inevitably they spend a lot of time thinking and talking about the features that distinguish the work of one artist from another. but those aren't what make artists good. what makes the line of a woman's cheek in a leonardo drawing good is how good it looks as the line of a cheek, not how little it looks like lines made by other artists.

because painting has such prestige, the myth that having a distinctive style (rather than painting well) is the defining quality of great artists has in turn given cover to a lot of bad design in adjacent fields. a brand that does something hideous to distinguish their products can say "like all great works of art, ours have a distinctive style," and people will buy it.

5 an ad that patek philippe ran in america in 1970 famously described a patek 3548 with a gold bracelet as a "$1700 trust fund." was it actually a good investment? in the very best case a dealer might pay you $20k now for one in unworn condition with its original box and papers. that's about a 4.5% rate of return, which is not absolutely terrible. but apparently the average rate of return on s&p 500 stocks over this period was more like 10%, if you reinvested all the dividends after paying taxes on them. the average rate of return would have been over 9% if you merely bought a lump of gold that hadn't been made into a watch. so, not surprisingly, the ad wasn't very good investment advice.

6 tania edwards, who ran us marketing for patek philippe in the 90s, said that bittel literally sketched the design of the 3919 on a piece of paper. this sounds odd to me, because the 3919 looked exactly like the existing 3520 with the addition of sub seconds (a small dial with a second hand above 6 o'clock). why would you sketch a design almost identical to an existing watch when you could just point to the existing watch and say "that, with sub seconds." what this story does show, though, is the degree to which people within patek felt their ad agency was responsible for the design of the 3919.

7 if i had to date the turning point for mechanical watches precisely, i'd say 1986. unit sales of swiss watches rebounded in 1985, but revenue didn't, which means what we're seeing is the boom in cheap quartz swatches. indeed, sales of mechanical watches must have been down if revenue was flat despite the sale of all those

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